

SECTION A: MULTIPLE CHOICE QUESTIONS - 20 MARKS

READ THE FOLLOWING INSTRUCTIONS

1. YOU HAVE TO START WITH SECTION A, AS YOU HAVE TO HAND IN THE OPTIC READER FORM AFTER 60 MINUTES.
2. As you may not erase wrong answers on the optic reader form, first circle your answers on this paper and only mark the answers on the form when you are certain of your choice.
3. Do all scribbling on the facing page. It will not be marked.
4. Use **SIDE 2** of the optic reader form.
Use a soft pencil.
Fill in your personal information in pencil on side 2.
Fill in your student number from top to bottom and then code it.
Answer questions 1 to 20 on the optic reader form on **SIDE 2**.
5. You may start with section B as soon as you have marked your answers on the optic reader form.

AFDELING A : MEERVOUDIGE KEUSEVRAE - 20 PUNTE

LEES DIE VOLGENDE INSTRUKSIES

1. U MOET MET AFDELING A BEGIN WANT U MOET DIE MERKLEESVORM NA 60 MINUTE INHANDIG.
2. U mag nie verkeerde antwoorde uitvee op die merkleesvorm nie, omkring dus u antwoorde eers in hierdie vraestel en merk die antwoorde op die vorm as u seker is van u keuse.
3. Doen alle krapwerk op die teenblad. Dit word nie nagesien nie.
4. Gebruik **KANT 2** van die merkleesvorm.
Gebruik 'n sagte potlood.
Vul u persoonlike inligting in potlood in op kant 2.
Vul u studentnommer van bo na onder in en kodeer dit.
Beantwoord vrae 1 tot 20 op die merkleesvorm se **KANT 2**.
5. U kan met afdeling B begin sodra u die antwoorde op die merkleesvorm ingevul het.

Question 1 / Vraag 1

$$y = \arctan(-1) \Rightarrow \tan y = -1 \text{ and } y \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

$$\arctan(-1) = \text{ / bgtan}(-1) =$$

[1a] $\frac{3\pi}{4} + 2n\pi$ or / of $\frac{7\pi}{4} + 2n\pi, n \in \mathbb{Z}$	[1b] $-\frac{\pi}{4} + n\pi, n \in \mathbb{Z}$	[1c] $\frac{7\pi}{4}$	[1d] $\frac{3\pi}{4}$
[1e] None of these / Geen van hierdie		☒ [1f] $-\frac{\pi}{4}$	[1g] $-\frac{5\pi}{4}$

Question 2 / Vraag 2

The domain of the function $f(x) = \arcsin(3 + \frac{x}{2})$ is

$$-1 \leq 3 + \frac{x}{2} \leq 1$$

Die definisieversameling van die funksie $f(x) = \arcsin(3 + \frac{x}{2})$ is

$$\Rightarrow -4 \leq \frac{x}{2} \leq -2$$

$$\Rightarrow -8 \leq x \leq -4$$

[2a] $(-\infty, \infty)$	[2b] $[-1, 1]$	[2c] $[-\frac{\pi}{2}, \frac{\pi}{2}]$	☒ [2d] $[-8, -4]$	[2e] $[-6 - \pi, -6 + \pi]$
[2f] $[\frac{5}{2}, \frac{7}{2}]$	[2g] $[3 - \frac{\pi}{4}, 3 + \frac{\pi}{4}]$	[2h] None of these / Geen van hierdie		

Question 3 / Vraag 3

The equation(s) of the vertical asymptote(s) of the function $f(x) = \frac{3x^2 - 5x + 2}{x^2 - 3x + 2}$ is (are)

Die vergelyking(s) van die vertikale asimptoot(tote) van die funksie $f(x) = \frac{3x^2 - 5x + 2}{x^2 - 3x + 2}$ is

[3a] $x = 1$ and / en $x = 2$	[3b] $x = 1$	[3c] $x = 2$	[3d] $x = 3$
[3e] $y = 1$ and / en $y = 2$	[3f] $y = 1$	[3g] $y = 2$	[3h] $y = 3$
[3i] None of these / Geen van hierdie			

$$f(x) = \frac{(3x-2)(x-1)}{(x-2)(x-1)}$$

$$= \frac{3x-2}{x-2}$$

Question 4 / Vraag 4

The equation(s) of the horizontal asymptote(s) of the function $f(x) = \frac{2x^2 + x}{x^2 - x}$ is (are)

Die vergelyking(s) van die horisontale asimptoot(tote) van die funksie $f(x) = \frac{2x^2 + x}{x^2 - x}$ is

[4a] $x = 0$ and / en $x = 1$	[4b] $x = 0$	[4c] $x = 1$	[4d] $x = 2$
[4e] $y = 0$ and / en $y = 1$	[4f] $y = 0$	[4g] $y = 1$	[4h] $y = 2$
[4i] None of these / Geen van hierdie			

$$f(x) = \frac{2x+1}{1-x}$$

In Questions 5 to 8 you have to calculate limits. If the answer is ∞ or $-\infty$ choose that and not "DNE". "DNE" means the limit does not exist.

In Vrae 5 tot 8 moet u limiete bereken. Indien die antwoord ∞ or $-\infty$ is, kies dit en nie "BN" nie. "BN" beteken dat die limiet nie bestaan nie.

Question 5 / Vraag 5

$$\lim_{x \rightarrow \infty} \frac{4x^2 + 4x + 1}{2x^4 - 7x + 4} = \lim_{x \rightarrow \infty} \frac{4x^2 + 4x^3 + 1x^4}{2x^4 - 7x^3 + 4x^4} = \frac{0}{2} = 0$$

[5a] $= \infty$	[5b] $= -\infty$	[5c] $= 0$	[5d] $= 2$	[5e] $= 0.25$	[5f] DNE / BN
[5g] None of these / Geen van hierdie					

Question 6 / Vraag 6

$$\lim_{x \rightarrow 2^+} \frac{x-3}{x^2-4} = \lim_{x \rightarrow 2^+} \frac{x-3}{(x-2)(x+2)} = -\infty$$

[6a] $= \infty$	[6b] $= -\infty$	[6c] $= 0$	[6d] $= 1$	[6e] $= 0.75$	[6f] DNE / BN
[6g] None of these / Geen van hierdie					

Question 7 / Vraag 7

$$\lim_{x \rightarrow 0^+} \frac{x}{\ln x}$$

[7a] $= \infty$	[7b] $= -\infty$	[7c] $= 0$	[7d] $= 1$	[7e] DNE / BN
[7f] None of these / Geen van hierdie				

Question 8 / Vraag 8

$$\lim_{x \rightarrow \infty} e^{0.3x} = \infty$$

[8a] $= \infty$	[8b] $= -\infty$	[8c] $= 0$	[8d] $= 1$	[8e] DNE / BN
[8f] None of these / Geen van hierdie				

Question 9 / Vraag 9

The equation of the tangent line to the graph of $y = e^{2x}$ at the point where $x = 0$ is

Die vergelyking van die raaklyn aan die grafiek van $y = e^{2x}$ in die punt waar $x = 0$ is

[9a] $y = (2e^{2x})x + 1$	[9b] $y = (2e^{2x})x$	[9c] $y = 2x$
[9d] $y = x$	[9e] $y = 2x + 1$	[9f] $y = x + 1$
[9g] None of these / Geen van hierdie		

$$\begin{aligned} f(x) &= e^{2x} \Rightarrow f(0) = 1 \\ f'(x) &= 2e^{2x} \Rightarrow f'(0) = 2 \\ y - 1 &= 2(x - 0) \\ \Rightarrow y &= 2x + 1 \end{aligned}$$

Question 10 / Vraag 10

The largest interval on which the function $f(x) = x\sqrt{x+4}$ is continuous, is

Die grootste interval waarop die funksie $f(x) = x\sqrt{x+4}$ kontinuu is, is

[10a] $[0, \infty)$	[10b] $(0, \infty)$	[10c] $[-4, \infty)$	[10d] $(-4, \infty)$	[10e] $(-\infty, \infty)$
[10f] None of these / Geen van hierdie				

$$\begin{aligned} D_f &= \{x | x+4 \geq 0\} \\ &= \{x | x \geq -4\} \end{aligned}$$

Question 11 / Vraag 11

The largest interval on which the function $f(x) = x\sqrt{x+4}$ is differentiable, is

Die grootste interval waarop die funksie $f(x) = x\sqrt{x+4}$ differensieerbaar is, is

[11a] $[0, \infty)$	[11b] $(0, \infty)$	[11c] $[-4, \infty)$	[11d] $(-4, \infty)$	[11e] $(-\infty, \infty)$
[11f] None of these / Geen van hierdie				

$$\begin{aligned} f'(x) &= \sqrt{x+4} + x \cdot \frac{1}{2} (x+4)^{-\frac{1}{2}} \\ &= \sqrt{x+4} + \frac{x}{2\sqrt{x+4}} \end{aligned}$$

Question 12 / Vraag 12

If / As $f(x) = \sin 2x$, then / dan is $f'(\frac{\pi}{6}) =$

$$\begin{aligned} f'(x) &= 2 \cos 2x \\ f'(\frac{\pi}{6}) &= 2 \cos 2 \cdot \frac{\pi}{6} = 2 \cos \frac{\pi}{3} = 2 \cdot \frac{1}{2} = 1 \end{aligned}$$

[12a] 1	[12b] $\frac{1}{2}$	[12c] $\sqrt{3}$	[12d] $\frac{\sqrt{3}}{2}$	[12e] None of these / Geen van hierdie
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Question 13 / Vraag 13

If $y^3 \sin x + \cos y = x$ then $\frac{dy}{dx} =$

As $y^3 \sin x + \cos y = x$ dan is $\frac{dy}{dx} =$

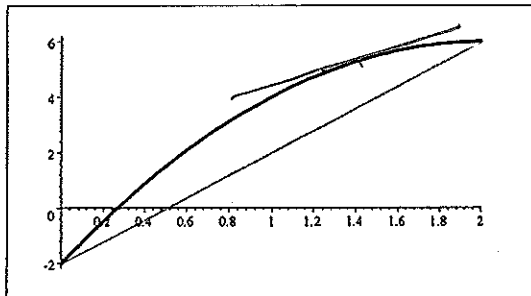
$$\begin{aligned} 3y^2 y' \sin x + y^3 \cos x - \sin y y' &= 1 \\ y' (3y^2 \sin x - \sin y) &= 1 - y^3 \cos x \end{aligned}$$

[13a] $\frac{1 - y^3 \cos x}{3y^2 \sin x - \sin y}$	[13b] $\frac{1}{3y^2 \cos x + \cos y}$	[13c] $\frac{1 - y^3 \cos x}{3y^2 \sin x}$
[13d] None of these / Geen van hierdie		

Question 14 / Vraag 14

Consider the graph of a function f below.

Beskou die grafiek van 'n funksie f hieronder.



• Slope of secant line is 4
• $f'(1.5)$ is the slope of the tangent line.

Which one of the statements below is true?

Watter een van die bewerings hieronder is waar?

Hint / Wenk: $\frac{f(2) - f(0)}{2 - 0} = \frac{6 - (-2)}{2} = 4$

[14a] $f'(1.5) > 4$ [14b] $f'(1.5) < 4$ [14c] $f'(1.5) = 4$

Question 15 / Vraag 15

Consider the statements:

Statement I: If $\lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$ exists, then f is differentiable at $x = a$. ✓

Statement II: If $\lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$ exists, then f is continuous at $x = a$. ✓

Which statement(s) is(are) true?

Beskou die bewerings:

Bewering I: As $\lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$ bestaan, dan is f differensieerbaar in $x = a$.

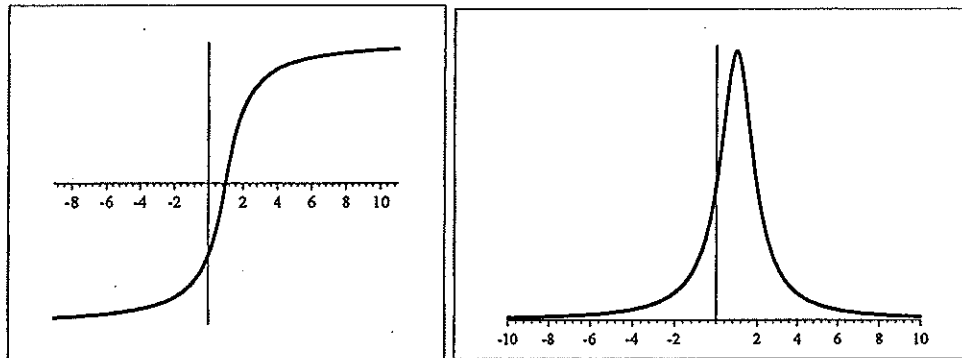
Bewering II: As $\lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$ bestaan, dan is f kontinu in $x = a$.

Watter bewering(s) is waar?

[15a] Only I / Slegs I	[15b] Only II / Slegs II
[15c] I and II / I en II	[15d] None of these / Geen van hierdie

Question 16 / Vraag 16

Consider the graphs below. / Beskou die grafieke hieronder.



$$y = f(x)$$

$$y = g(x)$$

Which statement is true? / Watter bewering is waar?

- ☒ [16a] $f'(x) = g(x)$ ☐ [16b] $g'(x) = f(x)$ ☐ [16c] $f'(x) = g'(x)$

Question 17 / Vraag 17

If $f(x) = x^3 + 6x^2 + 9x + 1$ then $f'(x) = 3x^2 + 12x + 9 = 3(x+3)(x+1)$.

$$f(0) = 1$$

On the closed interval $[-2, 0]$ the function f has an absolute maximum at $x =$

$$f(-2) = -8 + 24 - 18 + 1$$

As $f(x) = x^3 + 6x^2 + 9x + 1$ dan is $f'(x) = 3x^2 + 12x + 9 = 3(x+3)(x+1)$.

$$= -26 + 25 = -1$$

Op die geslote interval $[-2, 0]$ het die funksie 'n absolute maksimum by $x =$

$$f(-1) = -1 + 6 - 9 + 1$$

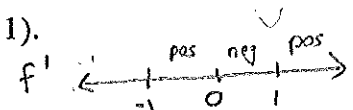
$$= -3$$

- ☒ [17a] 0 ☐ [17b] -1 ☐ [17c] -2 ☐ [17d] None of these / Geen van hierdie

Question 18 / Vraag 18

If $f(x) = 4x^6 - 6x^4 + 10$ then $f'(x) = 24x^5 - 24x^3 = 24x^3(x-1)(x+1)$.

At $x = 1$ the function f has



As $f(x) = 4x^6 - 6x^4 + 10$ dan is $f'(x) = 24x^5 - 24x^3 = 24x^3(x-1)(x+1)$.

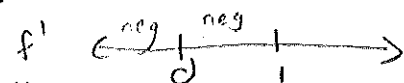
By $x = 1$ het die funksie

- ☐ [18a] a local maximum / 'n lokale maksimum ☒ [18b] a local minimum / 'n lokale minimum
☐ [18c] None of these / Geen van hierdie

Question 19 / Vraag 19

If $f(x) = 6x^4 - 8x^3 + 3$ then $f'(x) = 24x^3 - 24x^2 = 24x^2(x-1)$.

At $x = 0$ the function f has



As $f(x) = 6x^4 - 8x^3 + 3$ dan is $f'(x) = 24x^3 - 24x^2 = 24x^2(x-1)$.

By $x = 0$ het die funksie

- ☐ [19a] a local maximum / 'n lokale maksimum ☐ [19b] a local minimum / 'n lokale minimum
☒ [19c] None of these / Geen van hierdie

Question 20 / Vraag 20

Consider a function f that is continuous on $\mathbb{R}$ with
 $f(-4) = 5$, $f'(-4) = 0$ and $f''(-4) = 8$.

At $x = -4$ the function f has

Beskou 'n funksie f wat kontinu is op $\mathbb{R}$ met
 $f(-4) = 5$, $f'(-4) = 0$ and $f''(-4) = 8$.

By $x = -4$ het die funksie

[20a] a local maximum / 'n lokale maksimum	☒ [20b] a local minimum / 'n lokale minimum
[20c] I need more information to know whether the function has a local extreme at $x = -4$.	
Ek het meer inligting nodig om te besluit of die funksie 'n lokale ekstreem by $x = -4$ het.	

[20]

SECTION B: 30 MARKS

READ THE FOLLOWING INSTRUCTIONS

1. Answer all the following questions on this paper in ink.
2. Work done in pencil or in red ink will not be marked.
3. You are not allowed to use correction fluid ("Tipp-Ex").
4. Do all scribbling on the facing page. It will not be marked.
5. If you need more space for an answer, use the facing page and indicate it clearly by framing it.
6. Show all steps and calculations and explain your answers.

AFDELING B: 30 PUNTE

LEES DIE VOLGENDE INSTRUKSIES

1. Beantwoord die volgende vrae op hierdie vraestel in ink.
2. Werk wat in potlood of rooi ink beantwoord is word nie nagesien nie.
3. U mag nie korreger-vloeistof ("Tipp-Ex") gebruik nie.
4. Doen alle krapwerk op die teenblad. Dit word nie nagesien nie.
5. As u nog ruimte vir 'n antwoord nodig het, gebruik die teenblad en dui dit duidelik aan deur 'n raam daarom te trek.
6. Toon alle stappe en berekeninge en verduidelik u antwoorde.

Question 21 / Vraag 21

Find the derivative of the function. Don't simplify your answer.

Bepaal die afgeleide van die funksie. Moenie u antwoord vereenvoudig nie.

i $y = \frac{\tan(x)}{1-x^2}$ (Use the quotient rule / Gebruik die kwosiëntreël)

$$\frac{dy}{dx} = \frac{\sec^2 x (1-x^2) - \tan x (-2x)}{(1-x^2)^2}$$

[3]

ii $f(t) = \sqrt{\arcsin(t^3)}$ / $f(t) = \sqrt{\arcsin(t^3)}$

$$f'(t) = \frac{1}{2} (\arcsin t^3)^{-\frac{1}{2}} \times \frac{3t^2}{\sqrt{1-(t^3)^2}}$$

[2]

iii $f(x) = \arctan \sqrt{1-x}$ / $f(x) = \arctan \sqrt{1-x}$

$$f'(x) = \frac{1}{1+(\sqrt{1-x})^2} \times \frac{1}{2} (1-x)^{-\frac{1}{2}} (-1)$$

[2]

Question 22 / Vraag 22

Calculate the following limits.

Do not use the rule of l'Hospital.

Show your calculations.

If you write down only an answer you will get no marks for the answer.

Bepaal die volgende limiete.

Moenie die reël van l'Hospital gebruik nie.

Toon u berekeninge.

Indien u slegs 'n antwoord neerskryf sal u geen punte vir die antwoord kry nie.

i $\lim_{x \rightarrow -\infty} \frac{2x^5 + 3x^2 - 4x}{7x^3 - 8x + 22}$

$$= \lim_{x \rightarrow -\infty} \frac{2x^2 + 3/x - 4/x^2}{7 - 8/x^2 + 22/x^3}$$

$$= \infty$$

[2]

ii $\lim_{x \rightarrow 1^+} \left(\frac{1}{|x-1|} - \frac{3}{x-1} \right)$

$$= \lim_{x \rightarrow 1^+} \left(\frac{1}{x-1} - \frac{3}{x-1} \right)$$

$$= \lim_{x \rightarrow 1^+} \frac{-2}{x-1}$$

$$= -\infty$$

$$|x-1| = \begin{cases} x-1 & \text{if } x \geq 1 \\ -x+1 & \text{if } x < 1 \end{cases}$$

[2]

iii Calculate $\lim_{x \rightarrow 0} \frac{x}{\sqrt{x+1} - 1}$.

Do not use the rule of l'Hospital.

Show your calculations.

If you write down only an answer you will get no marks for the answer.

Bepaal $\lim_{x \rightarrow 0} \frac{x}{\sqrt{x+1} - 1}$.

Moenie die reël van l'Hospital gebruik nie.

Toon u berekeninge.

Indien u slegs 'n antwoord neerskryf sal u geen punte vir die antwoord kry nie.

$$\lim_{x \rightarrow 0} \frac{x}{\sqrt{x+1} - 1}$$

$$= \lim_{x \rightarrow 0} \frac{x}{\sqrt{x+1} - 1} \times \frac{\sqrt{x+1} + 1}{\sqrt{x+1} + 1}$$

$$= \lim_{x \rightarrow 0} \frac{x(\sqrt{x+1} + 1)}{(\sqrt{x+1} - 1)(\sqrt{x+1} + 1)}$$

$$= \lim_{x \rightarrow 0} (\sqrt{x+1} + 1)$$

$$= 2$$

[3]

Question 23 / Vraag 23

Consider the function / Beskou die funksie

$$f(x) = \begin{cases} x + a^2 & \text{if / as } x < 0 \\ \cos x & \text{if / as } x = 0 \\ x^2 + a & \text{if / as } x > 0 \end{cases}$$

- i Find $\lim_{x \rightarrow 0^+} f(x)$. / Bepaal $\lim_{x \rightarrow 0^+} f(x)$.

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} (x^2 + a) = a$$

[2]

- ii Find $\lim_{x \rightarrow 0^-} f(x)$. / Bepaal $\lim_{x \rightarrow 0^-} f(x)$

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} (x + a^2) = a^2$$

[2]

- iii Find the value(s) of a for which $\lim_{x \rightarrow 0} f(x)$ will exist..

Bepaal die waarde(s) van a waarvoor $\lim_{x \rightarrow 0} f(x)$ sal bestaan.

$$\lim_{x \rightarrow 0} f(x) \text{ exists} \Leftrightarrow \lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^+} f(x)$$

$$\Rightarrow a^2 = a$$

$$\Rightarrow a^2 - a = a(a - 1) = 0$$

$$\Rightarrow a = 0 \text{ or } a = 1$$

[2]

- iv Complete the definition: The function f is continuous at $x = 0$ if..

Voltooi die definisie: Die funksie f is kontinu in $x = 0$ as...

① $f(0)$ exists.

② $\lim_{x \rightarrow 0} f(x)$ exists

③ $\lim_{x \rightarrow 0} f(x) = f(0)$

[1]

- v Find the value(s) of a for which $y = f(x)$ will be continuous at $x = 0$.

Bepaal die waarde(s) van a waarvoor $y = f(x)$ kontinu in $x = 0$ sal wees.

$$f(0) = \cos 0 = 1$$

from iii

$$\text{if } a = 0 \text{ then } \lim_{x \rightarrow 0} f(x) = 0 \neq f(0)$$

$$\text{if } a = 1 \text{ then } \lim_{x \rightarrow 0} f(x) = 1 = f(0)$$

[2]

f is continuous at $x = 0$ if $\underline{a = 1}$

Question 24 / Vraag 24

Find the critical numbers of $f(x) = x^3 \ln x$.

Take care with the differentiation, because you will not get any marks for your answer if you make a mistake with the differentiation.

Bepaal die kritieke getalle van $f(x) = x^3 \ln x$.

Wees versigtig met die differensiasie, want u sal geen punte vir jou antwoord kry as die differensiasie verkeerd is.

$$\begin{aligned} f'(x) &= 3x^2 \ln x + x^3 \frac{1}{x} \\ &= 3x^2 \ln x + x^2 \\ &= x^2 (3 \ln x + 1) \end{aligned}$$

$$\begin{aligned} f'(x) = 0 &\Rightarrow x = 0 \text{ or } \ln x = -\frac{1}{3} \\ &\Rightarrow x = 0 \text{ or } x = e^{-\frac{1}{3}} \end{aligned}$$

but a critical number must be in
the domain

The only critical number is $c = e^{-\frac{1}{3}}$

[3]

Question 25 / Vraag 25

Find the interval(s) on which the function $f(x) = xe^{-x^2}$ is increasing.

Hint: $\frac{d}{dx}xe^{-x^2} = e^{-x^2} - 2x^2e^{-x^2} = e^{-x^2}(1 - 2x^2)$

Bepaal die interval(le) waarop die funksie $f(x) = xe^{-x^2}$ stygend is.

Wenk: $\frac{d}{dx}xe^{-x^2} = e^{-x^2} - 2x^2e^{-x^2} = e^{-x^2}(1 - 2x^2)$

$$f'(x) > 0 \Rightarrow e^{-x^2}(1 - 2x^2) > 0$$

$$\Rightarrow 1 - 2x^2 > 0 \text{ as } e^{-x^2} > 0 \text{ for all } x$$

$$\Rightarrow (1 - \sqrt{2}x)(1 + \sqrt{2}x) > 0$$

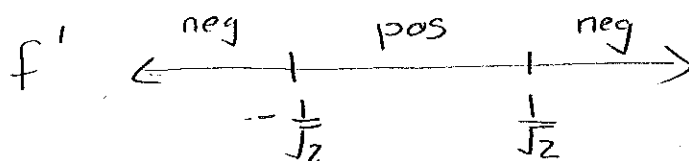
$$\Rightarrow x \in \left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$$

OR

$$f'(x) = 0 \Rightarrow e^{-x^2}(1 - 2x^2) = 0$$

$$\Rightarrow 1 - 2x^2 = 0$$

$$\Rightarrow x = \pm \sqrt{\frac{1}{2}} = \pm \frac{1}{\sqrt{2}}$$



f is increasing on $\left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$

[4]